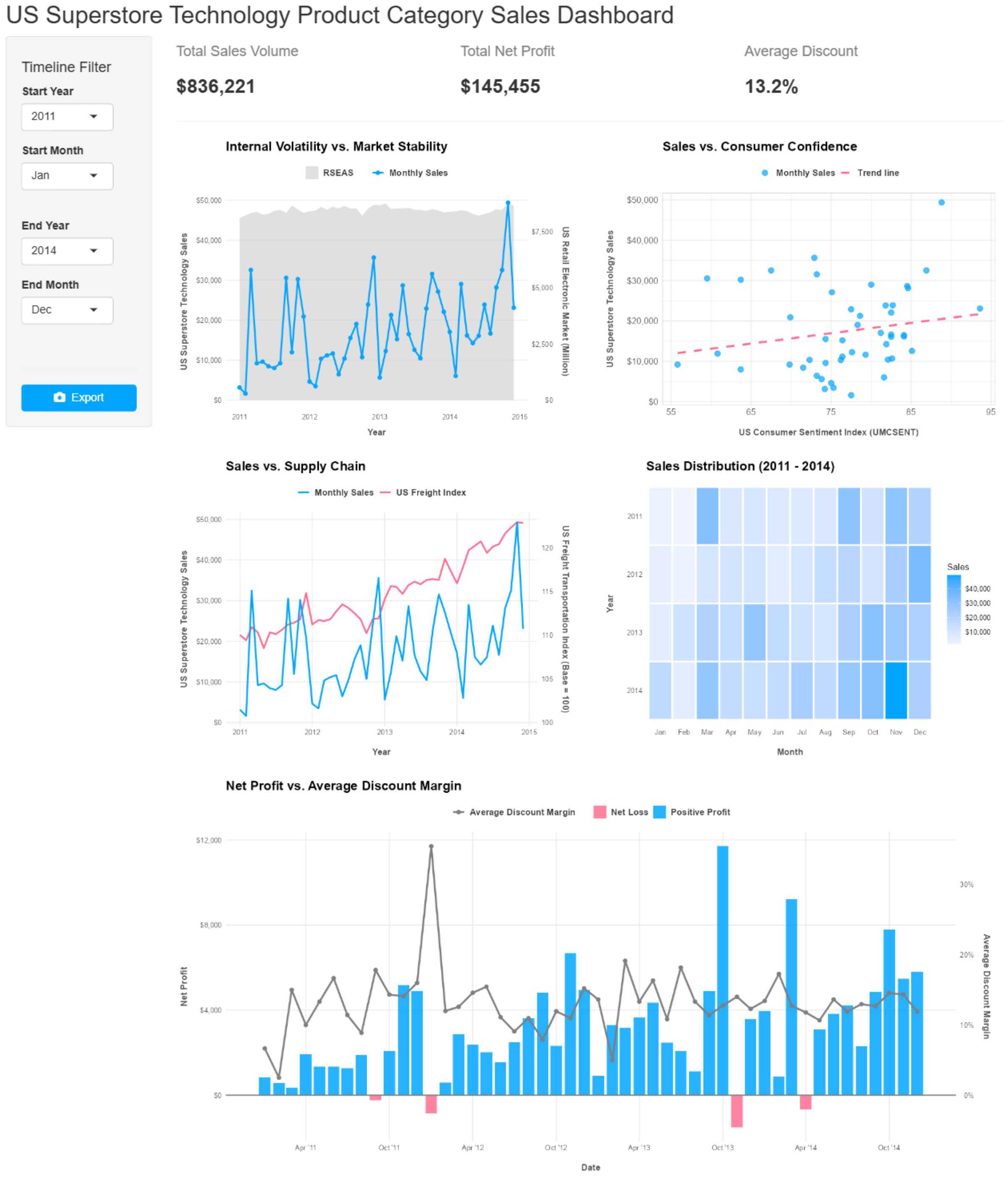
Dashboard


US Superstore Technology Product Category Sales Dashboard
Timeline Filter
Start Year
2011
Start Month
Jan
End Year
2014
End Month
Dec
Export
Total Sales Volume
$836,221
Total Net Profit
$145,455
Average Discount
13.2%
Internal Volatility vs. Market Stability
RSEAS
Monthly Sales
US Superstore Technology Sales
US Retail Electronic Market (Million)
Year
Sales vs. Consumer Confidence
Monthly Sales
Trend line
US Superstore Technology Sales
US Consumer Sentiment Index (UMCSENT)
Sales vs. Supply Chain
Monthly Sales
US Freight Index
US Superstore Technology Sales
US Freight Transportation Index (Base = 100)
Year
Sales Distribution (2011 - 2014)
Year
Month
Sales
Net Profit vs. Average Discount Margin
Average Discount Margin
Net Loss
Positive Profit
Net Profit
Average Discount Margin
Date

Problem Design

The business problem to investigate.

The Superstore's technology product category is experiencing severe, non-seasonal sales volatility characterized by unpredictable month-over-month fluctuations. This erratic perfomance points to systematic internal operational failure, which ultimately drives margin erosion and negatively impacts overall profitability.

The challenges to face.

What is the reason for the large fluctuations in sales? Can Superstore's management blame macroeconomic effects? If sales are not driven by external economics, is the root cause an internal operational issue?

Dashboard objective

The interactive R Shiny dashboard is designed as a direct strategic intervention to transition the organization from a reactive stance to a predictive stance. The tool achieves this by tracking the business through a dual-lens approach:

- By introducing reliable and authoritative external macroeconomic datasets to investigate and validate the correlation of past internal sales. This enables stakeholders to analyse the root causes of sales fluctuations and make quick decision.
- By filtering and analysing baseline sales metrics across a customizable timeline, the dashboard exposes the true, underlying consumer baseline. This visibility enables management to establish stable pricing structures, protect product value, and optimize inventory procurement.
- The interactive interface arms decision-makers with the visibility required to eliminate unpredictable, volume-chasing markdowns, ultimately halting self-inflicted sales decline and protecting long-term division profitability.

Dataset Selection

1. **Superstore Dataset**

   The Superstore dataset serves as the core internal performance baseline for this analysis. Filtered specifically to the US market (2011-2014), it provides granular transaction data, inclusing sales volume, shipping timelines, and profit margins. In this dashboard, this internal data acts as the dependent variable, allowing us to isolate the company's severe Q3/Q4 revenue volatility and contrast it against external macroeconomic and supply chain indicators.

2. **Advance Retail Sales: Electronics and Appliance Stores Dataset**

   - U.S. Census Bureau. *Advance Retail Sales: Electronics and Appliance Stores* [RSEAS]. FRED. https://fred.stlouisfed.org/series/RSEAS
   - This dataset measures national retail sales for electronic and appliance stores, providing macroeconomic baseline for technology consumer spending.
   - This dataset narrows down to the electronics sector and prevents skew from broad retail trends. While the national electronics market follows a stable, predictable growth trend with only mild seasonal fluctuations, the Superstore dataset suffers from violent, bi-annual whiplash. This massive year-round divergence proves the company's erratic cycle is a severe internal vulnerability, rather than a reflection of the broader industry.

3. **Consumer Sentiment Dataset**

   - University of Michigan. *University of Michigan: Consumer Sentiment* [UMCSENT]. FRED. https://fred.stlouisfed.org/series/UMCSENT
   - This dataset measures the economic confidence and purchasing willingness of US households.
   - This dataset shows whether the severe volatility of the Superstore dataset is driven by fluctuating macroeconomic anxiety or by independent seasonal behaviours. Overlaying this dataset reveals a lack of correlation, proving that extreme seasonal purchasing overrides general consumer sentiment. Therefore, severe sales volatility of the Superstore dataset may calendar-driven (e.g., holiday promotions, tax seasons) or other reasons but highly insulated from broader economic pessimism.

4. **Logistics and Supply Chain Dataset**

   - U.S. Bureau of Transportation Statistics. *Freight Transportation Services Index* [TSIFRGHT]. FRED. https://fred.stlouisfed.org/series/TSIFRGHT
   - The Transportation Services Index measures month-to-month volume changes in national shipping and freight across all transport modes.
   - This dataset was selected to determine if extreme sales volatility of the Superstore dataset is a byproduct of national supply chain collapses, or an internal operational failure. Comparing the datasets reveals a stark contrast: the national freight network grows steadily with moderate seasonality, while the Superstore datasets experiences violent bi-annual whiplash. This lack of correlation proves the Superstore erratic sales cycle is not caused by the US logistics network shutting down.

Visualizations and Interpretation

Chart 1

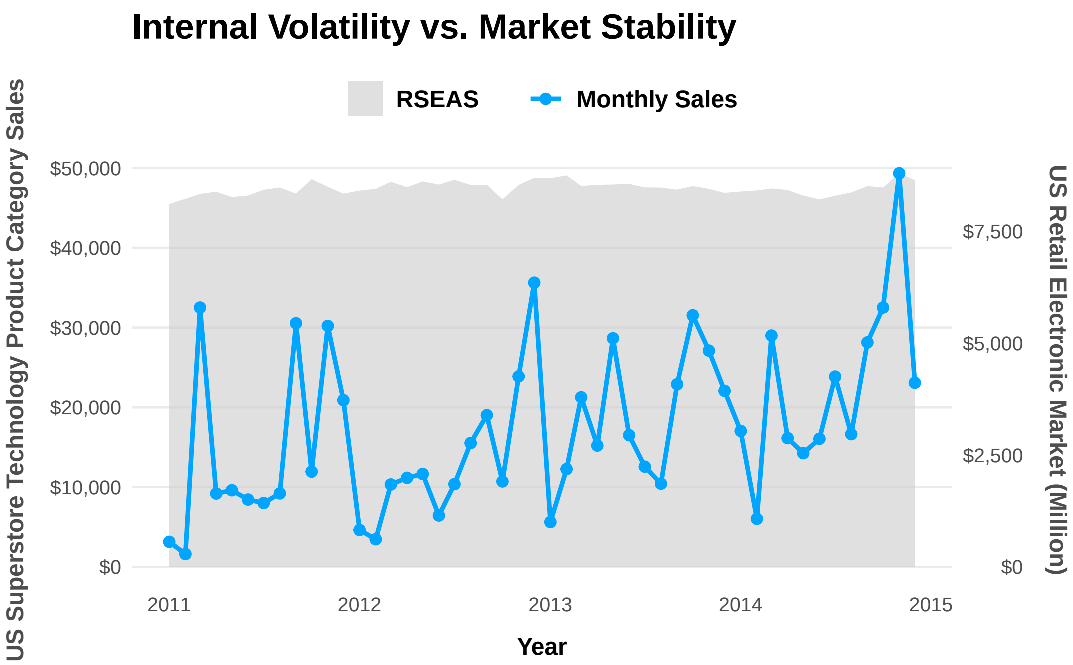


Chart 1 utilizes a line chart overlaid on an area graph to reveal a stark contrast between the smoothly growing national retail electronics market and the extreme, erratic spikes and crashes of internal technology product category sales. This massive divergence is occurring because internal sales volume has become completely detached from organic consumer demand, driven instead by localized, artificial events such as aggressive clearance sales and reactive promotions. Management should interpret this extreme sales volatility as entirely self-inflicted, meaning external retail downturns cannot be used as an excuse for poor performance when the macroeconomic baseline remains healthy and stable.

Academically, this reflects the theory of Demand Distortion and the Promotional Pulse Effect, where a company's own aggressive pricing strategies alter consumer buying cycles and destroy natural seasonality. The actionable business strategy requires leadership to acknowledge the absence of true organic seasonality, implement strict, data-driven forecasting to align inventory with national market trends, restrict margin destroying ad-hoc promotional events, and utilize the interactive dashboard to monitor baseline demand rather than relying on panic-selling.

Chart 2

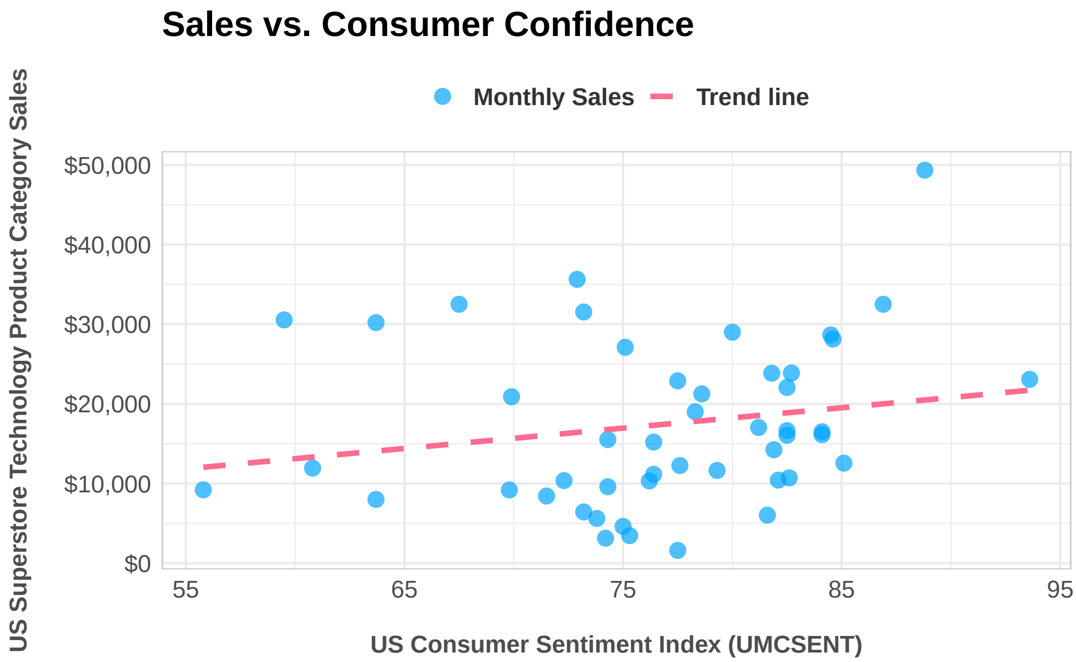


Chart 2 features a scatter plot demonstrating emonstrating a remarkably weak, shallow trendline between the US Consumer Sentiment Index and internal sales, which mathematically proves a lack of meaningful correlation between the two. This disconnect happens because consumers are purchasing strictly in response to deep, temporary price cuts rather than acting on their overall confidence in the broader economy, generating massive sales peaks even during months of high national economic anxiety. This should be interpreted as definitive proof that relying on "consumer economic anxiety" as a defense for low-performing months is statistically invalid; the business is artificially forcing sales through discounting rather than capturing organically confident buyers.

This is an example of the Price Elasticity of Demand, illustrating that the customer base is highly price sensitive and that the sheer magnitude of the discounts easily overpowers any broader macroeconomic sentiment (Wikipedia Contributors, 2019a). The actionable strategy demands that the company entirely abandon macroeconomic excuses, immediately reduce its reliance on deep discounting, retrain the customer base to purchase outside of promotional windows, and actively track price elasticity metrics to protect long-term profitability.

Chart 3

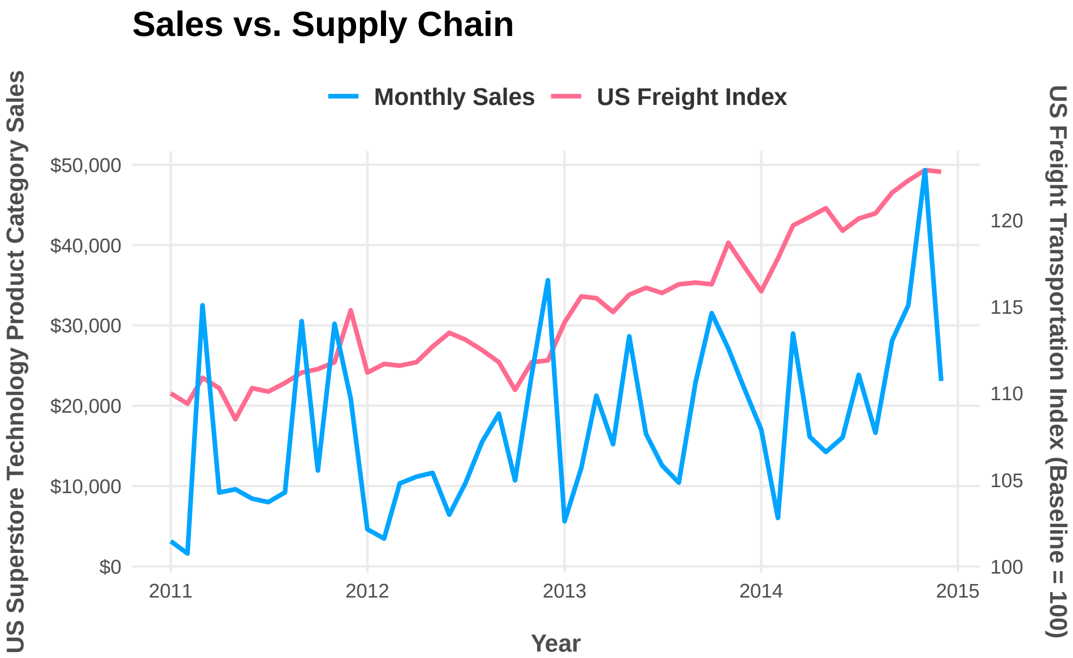


Chart 3 employs a dual-axis line chart to contrast the steady, predictable scaling of the national US Freight Transportation Index with the violent, erratic internal sales. This massive disconnect is happening because the business relies heavily on unpredictable promotional events rather than stable forecasting, which abruptly forces massive, unforecasted shipping volumes onto the logistics network at random intervals. We must interpret this to mean that any internal complaints regarding supply chain bottlenecks, delayed shipments, or exorbitant expedited freight costs are purely internally generated, as the national logistics network is actually functioning perfectly and scaling as expected.

This scenario perfectly illustrate the Bullwhip Effect in supply chain management, where artificially induced demand spikes at the retail level create severe, amplified inefficiencies and logistical strain upstream (Wikipedia Contributors, 2019b). The resulting actionable strategy requires the company to stop treating logistics issues as external failures and instead plan to stabilize these artificially induced demand shocks by aligning promotional calendar with standard logistical planning and forecasting freight needs based on stable organic demand.

Chart 4

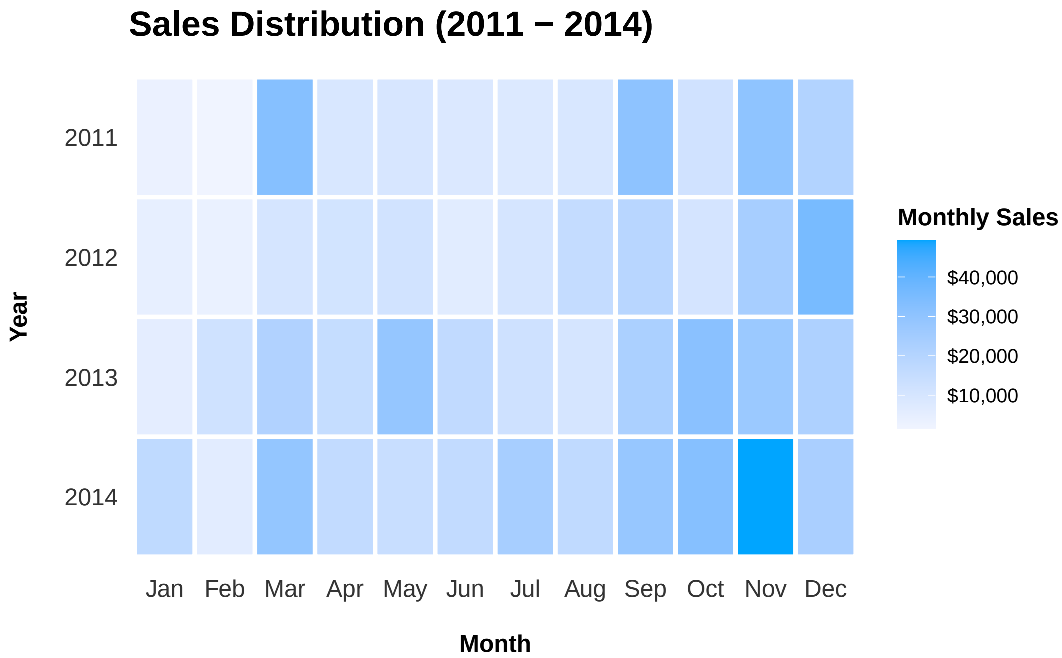


Chart 4 utilizes a heatmap to visualize the distribution of monthly sales across four years, revealing a scattered, "checkerboard" pattern of sales peaks rather than consistent vertical bands of color. This erratic distribution is happening because the company lacks genuine, organic recurring demand; instead, high-sales months are artificially generated by the timing of unpredictable, ad-hoc internal promotional events that shift completely from year to year. We must interpret this visual as definitive proof that the concept of natural seasonality within this business unit is a complete myth, explicitly dismantling any managerial claims that cyclical market forces dictate their sales perfomance.

This reflects the theory of the Promotional Pulse Effect again, where reactive discounting strategies entirely overwrite organic consumer purchasing rhythms, creating a completely artificially and chaotic demand schedule. The resulting actionable strategy requires leadership to cease planning around non-existent seasonal trends, restructure their marketing calendar to cultivate genuine baseline demand, and utilize the dashboard to enforce strict guardrails against randomly timed clearance events that continually disrupt operational predictability.

Chart 5

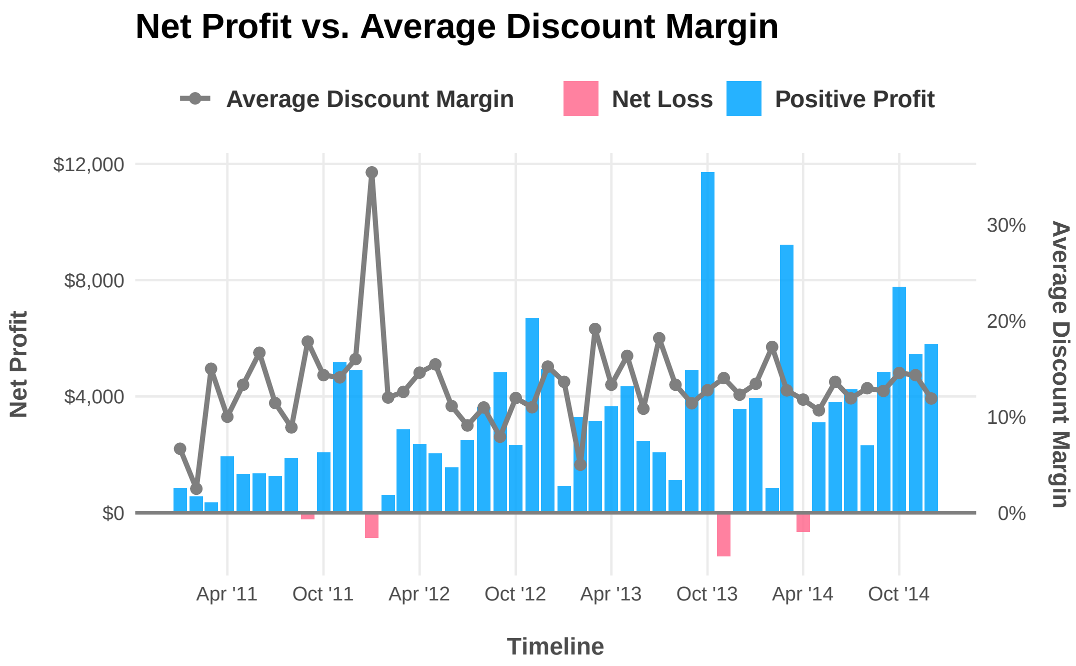


Chart 5 employs a dual-axis bar and line visualization to expose a critical operational flaw by directly contrasting monthly Net Profit (represented by the bars) against the Average Discount Margin (represented by the overlaid line). This visualization clearly demonstrates that whenever the average discount line spikes upwards to force a surge in sales volume, the corresponding net profit bar plummets, dropping below the zero baseline into negative territory (represented by the red bars). This margin destruction occurs because leadership is relying on severe price-slashing to clear unforecasted inventory and artificially inflate top-line revenue, effectively selling large volumes of technology products at a net financial loss to the business. Management must interpret this as absolute, undeniable proof that their current promotional strategy is financially unsustainable; driving high sales volume is utterly meaningless if the associated discounting actively destroys the bottom line.

This illustrates Reference Price Theory, which dictates that frequent, aggressive promotions lower the consumer's internal reference price, effectively training the market to wait for discounts and forcing the company into a continuous, destructive cycle of margin erosion just to maintain baseline volume (Bennett, 2023). The critical actionable strategy is to immediately halt ad-hoc clearance events and transition to the predictive interactive dashboard, allowing the company to forecast inventory accurately, protect baseline product value and permanently transition away from unprofitable, volume-chasing sales tactics.

Data Story

What is happening?

The Superstore's technology product category is experiencing severe, non-seasonal sales volatility characterized by unpredictable month-over-month fluctuations. This erratic perfomance points to systematic internal operational failure, which ultimately drives margin erosion and negatively impacts overall profitability.

Is it affected by external factors?

By analysing other external data, such as US Consumer Sentiment Index, US Freight Transportation Index and etc, it is not found this phenomenon can be attributed to he bad economy, supply chain bottleneck or a natural off-season for revenue issues. Thus, the data firmly disproves all of these external excuses.

From the economic perspective, our sales show zero correlation with the US Consumer Sentiment Index, meaning national macroeconomic anxiety is not dictating our sales drops.

The national freight network scales smoothly, this means that the broader US logistics and shipping industry is perfectly healthy. The shipping networks are growing and operating in a stable, predictable, and normal upward trend. In other words, there is no national "shipping crisis" happening. It is actually the Superstore's management unforecasted sales spikes that are forcing violent inefficient shipping bursts onto the logistics network. Therefore, instead of having steady, predictable sales, the Superstore's management runs massive, random clearance discounts that leads in a surge in sales. To fulfill all these sudden orders, the warehouse force massive volumes of boxes into the shipping network all at once. This unpredictable incident also resulted in a lack of time to order new stocks, resulting in a cliff-like drop in sales next month. Hence, overall sales have been on a rollercoaster ride.

A four-year heatmap also reveals no natural seasonal purchasing patterns, the Superstore's high-sales months are scattered randomly like a checkerboard.

Why is this happening?

Because external factors are not to blame, this volatility points directly to a systematic internal operational failure, especially a destructive cycle of panic-discounting. Due to a lack of true organic seasonality and struggles with inventory forecasting, sales are artificially forced by running massive and unpredictable clearance events. The timing of these extreme and reactive discounts completely overwrites natural consumer buying cycles, making internal promotions the sole driver of these chaotic sales spikes.

What is the financial impact?

This strategy is creating a discount trap that is actively bleeding the bottom line. When average discount margins are overlaid against net profit, the correlation is undeniable. Aggressive discounting successfully drives sales volume, but it plunges net margins directly into negative territory. During the highest-volume months, the discounts are so severe that the company is effectively paying customers to take inventory, driving significant margin erosion. Without intervention to change this strategy, this self-inflicted margin destruction will continue drain the company's profitability while hiding behind the illusion of high sales.

How should the business respond?

To stem the instability of sales, the company should immediately shift away from reactive discounting and embrace data-driven forecasting. The data suggests focusing on:

- Immediately restrict the randomly timed clearance events that continually disrupt operational predictability and destroy product value.
- Utilize the new interactive forecasting tool to track the true, organic baseline of consumer demand.
- Restructure inventory procurement and marketing calendars to align with stable marketing growth, permanently transitioning away from unprofitable, volume chasing sales strategies.

What happens if action is delayed?

If intervention is delayed, sales will continue to fluctuate wildly, which could lead to significant negative profits. Every month the company continues to rely on unforecasted clearance events, product value is further diminished, and the customer base is increasingly trained to only purchase at a steep discount. Delaying the transition to data-driven forecasting by even one quarter will result in hundreds of thousands of dollars in continued self-inflicted losses and further destabilization of the supply chain.

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I acknowledge the use of Gemini to help me with (a). searching related credible external datasets; (b). explaining business and economics terminology and related knowledge; (c). correcting R code for data visualisation; (d). generating an interactive dashboard based on my prompt; (e). rephrasing my explanation and justification to be more business-professional, including grammatical structures, punctuation and vocabulary; (f). summarising how to apply lecturer materials to this assignment to follow the assignment rubric.